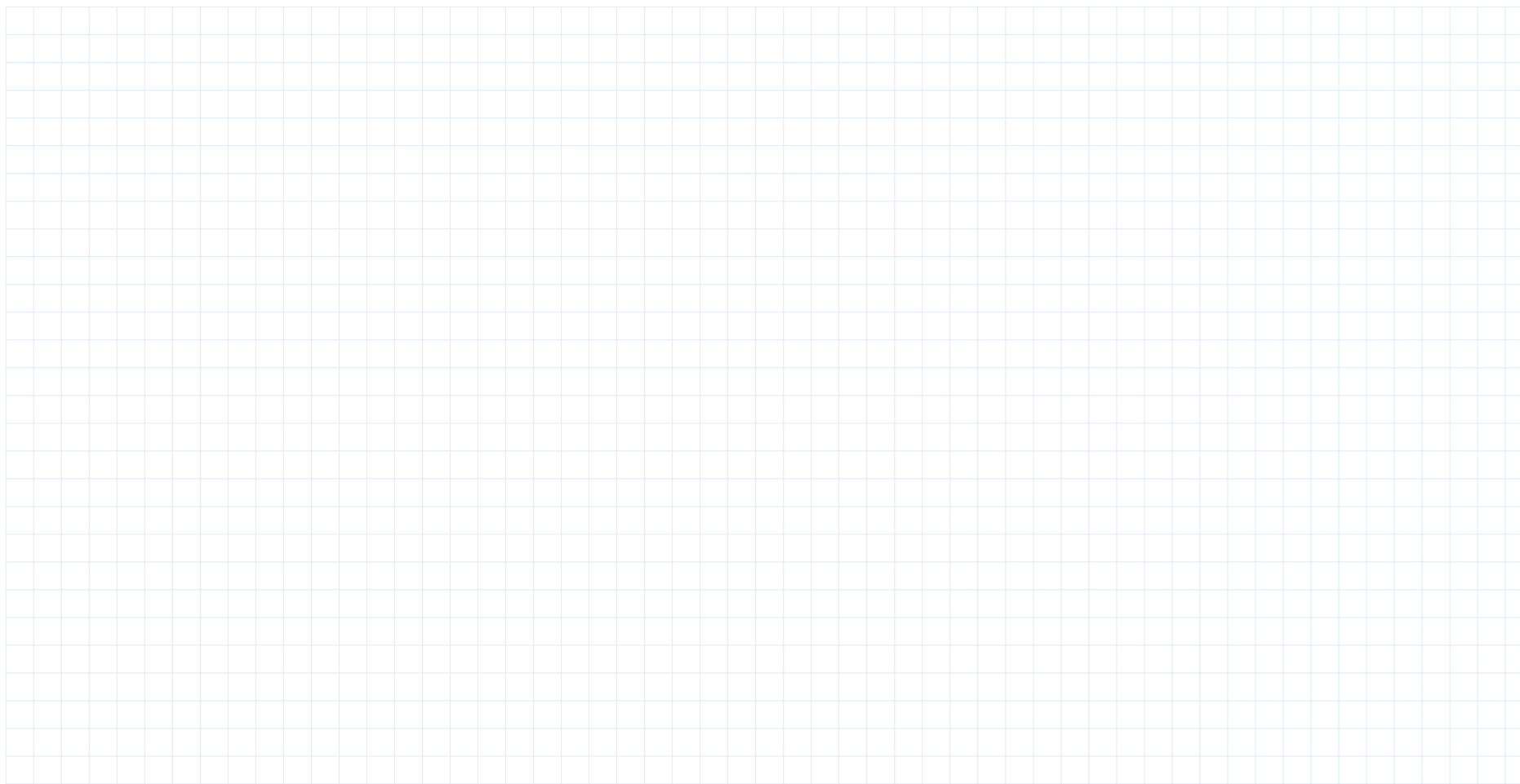


What Is an Identity?

An **identity** is true for every input for which both sides are defined. Establish an identity by rewriting one side until it equals the other. Prove the following using sine and cosine:

$$\sec x \sin x = \tan x.$$

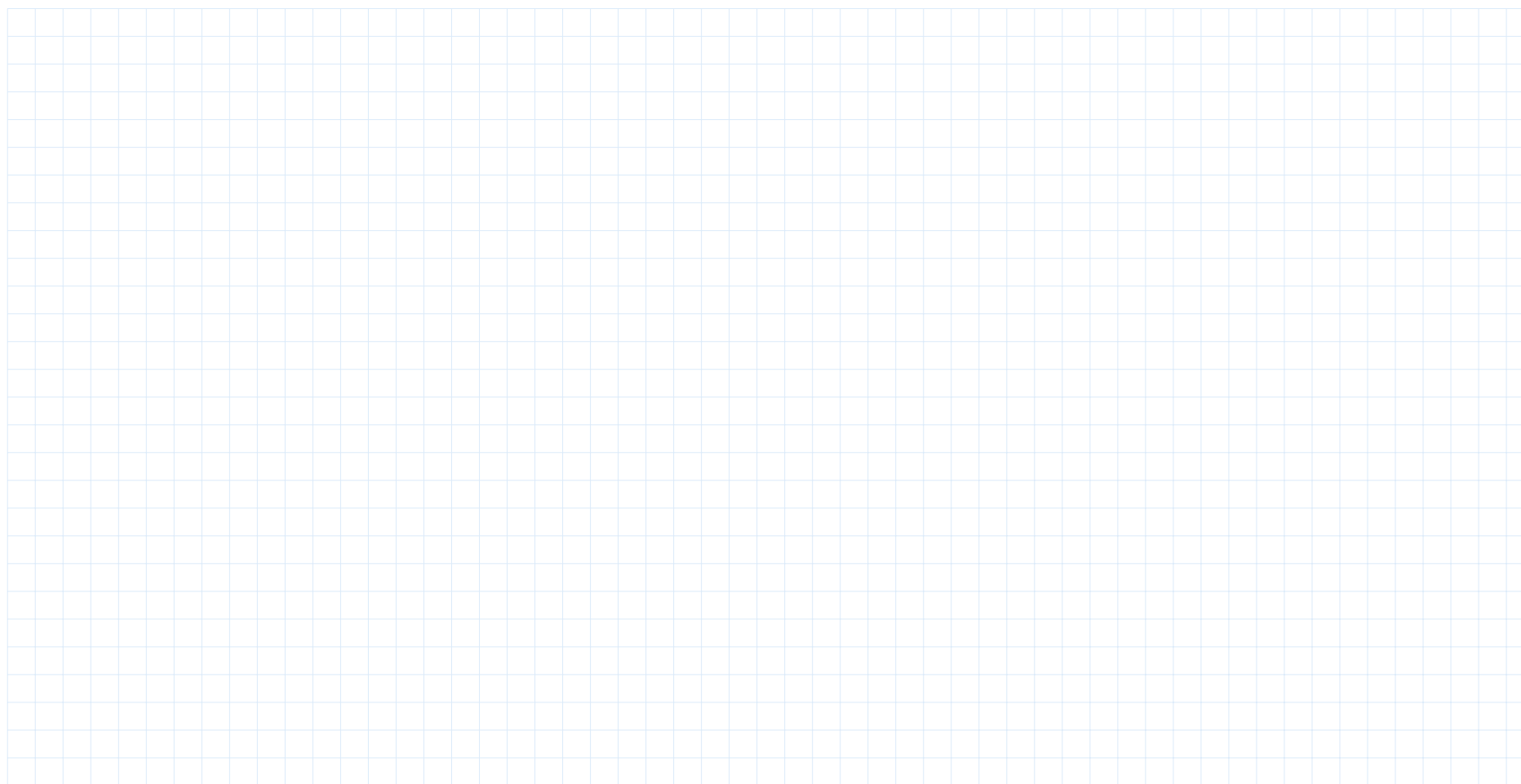


Simplify by Factoring

Use $1 - \cos^2 x = \sin^2 x$ to simplify

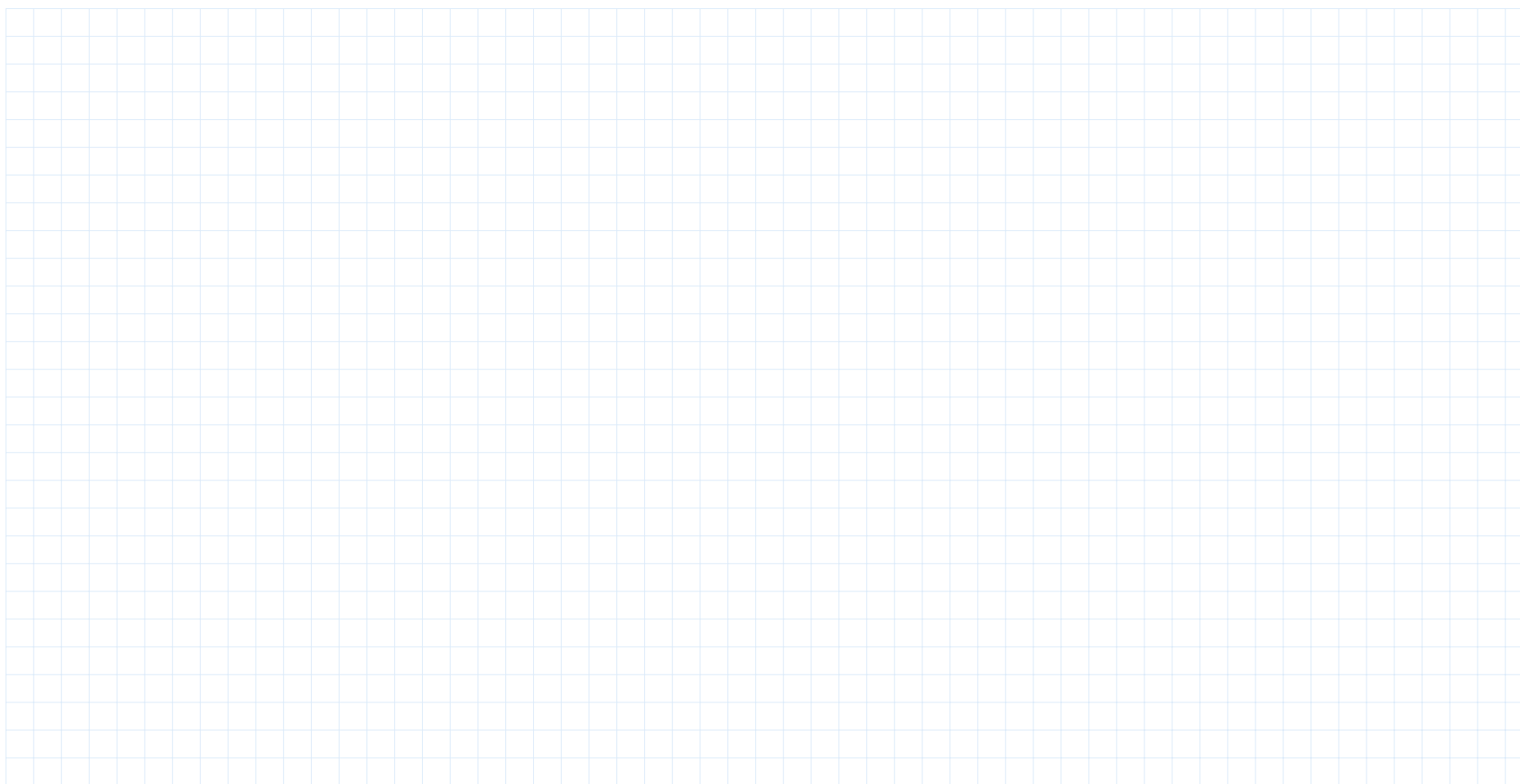
$$\frac{1 - \cos^2 x}{\sin x}.$$

Retain the restriction from the original denominator.



Establish the identity by using a common denominator:

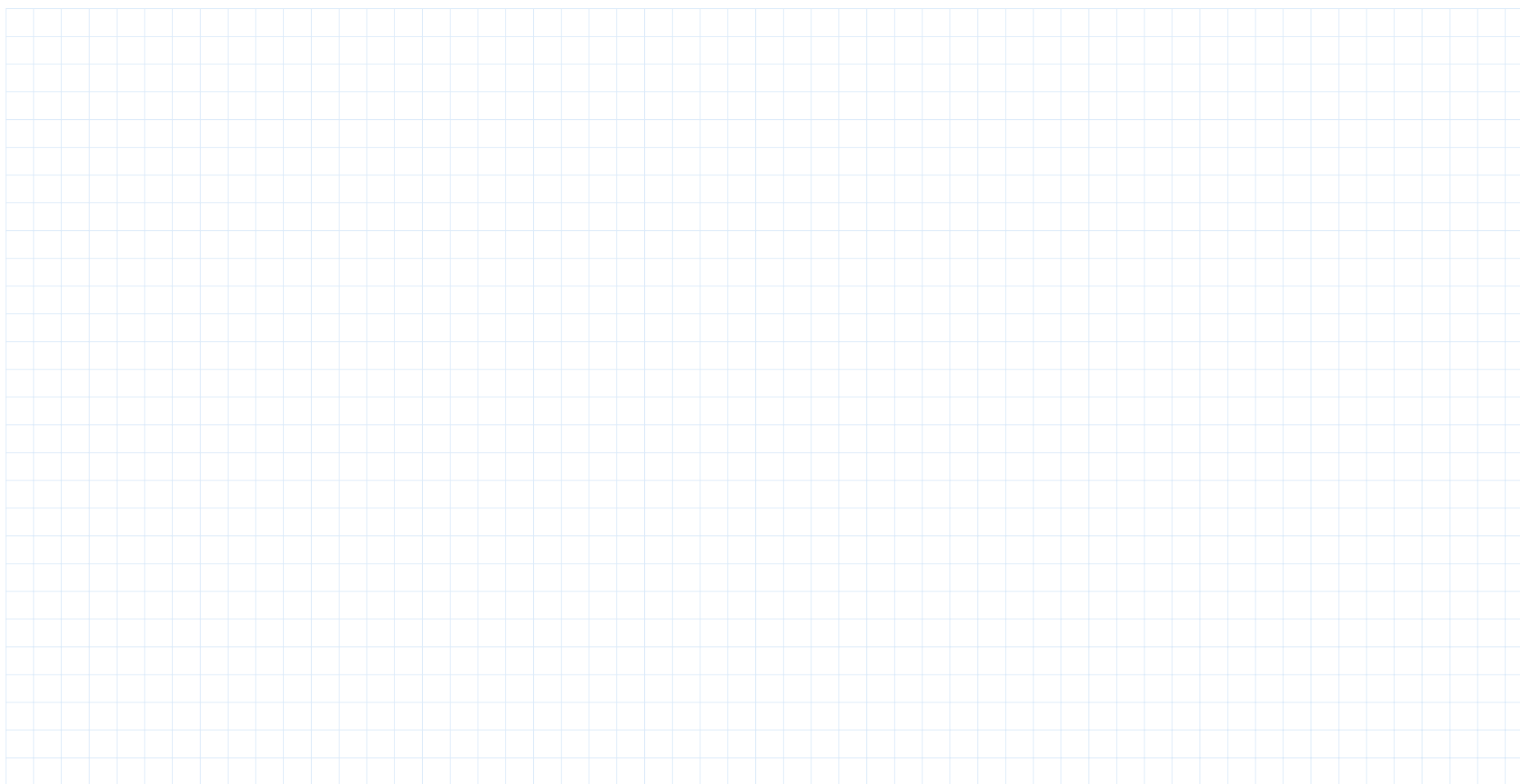
$$\frac{1}{1 - \sin x} + \frac{1}{1 + \sin x} = 2 \sec^2 x.$$



Multiply by a Conjugate

Multiplying by $\frac{1 + \cos x}{1 + \cos x}$ can reveal a Pythagorean identity. Establish, on the common domain,

$$\frac{\sin x}{1 - \cos x} = \frac{1 + \cos x}{\sin x}.$$



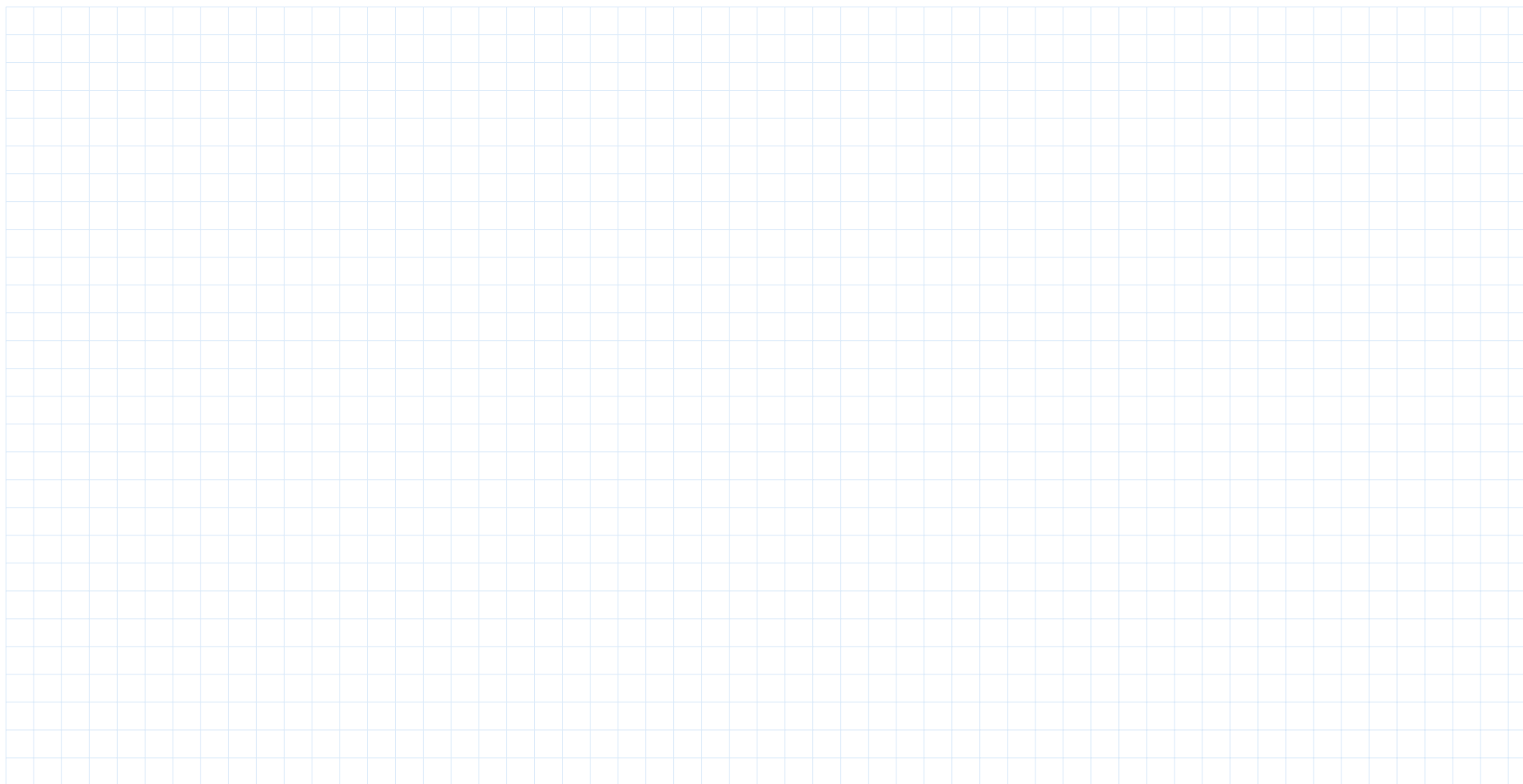
Even and Odd Identities

Section 6.4

Use $\sin(-x) = -\sin x$ and $\cos(-x) = \cos x$ to simplify

$$\frac{\sin^2(-x) + \cos^2(-x)}{\cos(-x)}.$$

State the restriction on the original expression.



A Factored Identity

Use $1 + \cot^2 x = \csc^2 x$ to establish the identity:

$$\frac{\cot^2 x}{\csc x - 1} = \csc x + 1.$$

Show the factorization that permits cancellation.

